

Question 1

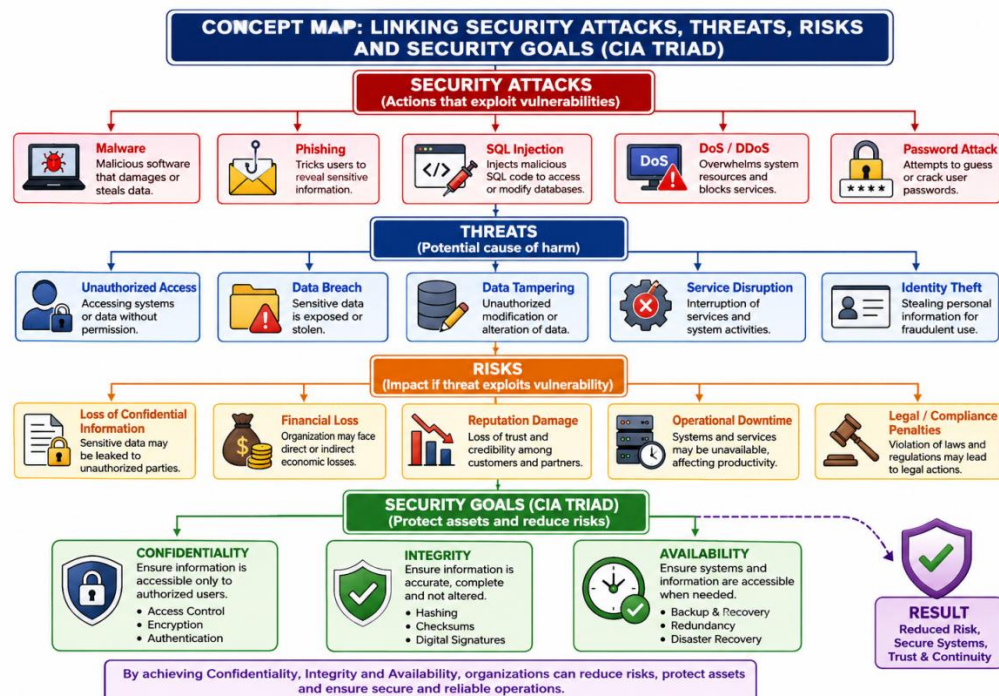
Create a Concept Map linking the following:

Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability)

Introduction

Information security aims to protect computer systems and data from unauthorized access, modification, and destruction. A security attack exploits system vulnerabilities, creating threats that may lead to risks. The CIA (Confidentiality, Integrity, and Availability) security goals help organizations minimize these risks and ensure secure communication.

Concept Map



Explanation of the Concept Map

1. Security Attacks

Security attacks are deliberate actions performed by attackers to compromise a computer system.

Examples include:

- Malware
- Phishing
- SQL Injection
- Denial of Service (DoS)

- Password Cracking

These attacks exploit weaknesses in systems.

2. Threats

A threat is the possibility that an attack can cause damage to an organization.

Examples:

- Unauthorized access
- Data tampering
- Identity theft
- Service disruption

Threats arise because of successful security attacks.

3. Risks

Risk is the potential loss resulting from a successful threat.

Common risks include:

- Data loss
- Financial loss
- Reputation damage
- Legal penalties
- Business interruption

Organizations evaluate these risks before implementing security controls.

4. Security Goals (CIA Triad)

The CIA Triad represents the three fundamental goals of information security.

Confidentiality

Ensures that information is accessible only to authorized users.

Examples:

- Password protection
- Encryption
- Access control

Integrity

Ensures that data remains accurate, complete, and unaltered.

Examples:

- Hash functions
- Digital signatures
- Checksums

Availability

Ensures that systems and services remain available whenever users need them.

Examples:

- Backup servers
- Disaster recovery
- Load balancing

Task (a): Show how each attack leads to specific threats and risks

Security Attack	Threat Created	Risk
Malware	Data corruption	Loss of important files
Phishing	Credential theft	Financial fraud
SQL Injection	Database modification	Data leakage
DoS Attack	Service interruption	Business downtime
Password Cracking	Unauthorized access	Privacy violation

Task (b): Map how security goals help mitigate them with one real-world example

Security Goal	Mitigates	Real-world Example
Confidentiality	Prevents unauthorized access	Internet banking uses OTP and encryption

Security Goal	Mitigates	Real-world Example
Integrity	Prevents data modification	Digital signatures in online documents
Availability	Prevents service downtime	Google Cloud backup servers maintain service availability

Analysis

Security attacks create threats, and threats increase organizational risks. The CIA security goals provide effective protection by preventing unauthorized access, ensuring data accuracy, and maintaining service availability. Implementing all three goals together significantly improves overall cybersecurity.

Conclusion

Security attacks, threats, and risks are interconnected components of cybersecurity. Organizations reduce these risks by implementing the CIA Triad—Confidentiality, Integrity, and Availability. A well-designed security framework protects sensitive information, maintains trust, and ensures continuous system operation.

Question 2

Develop a Concept Map Comparing Substitution Techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher

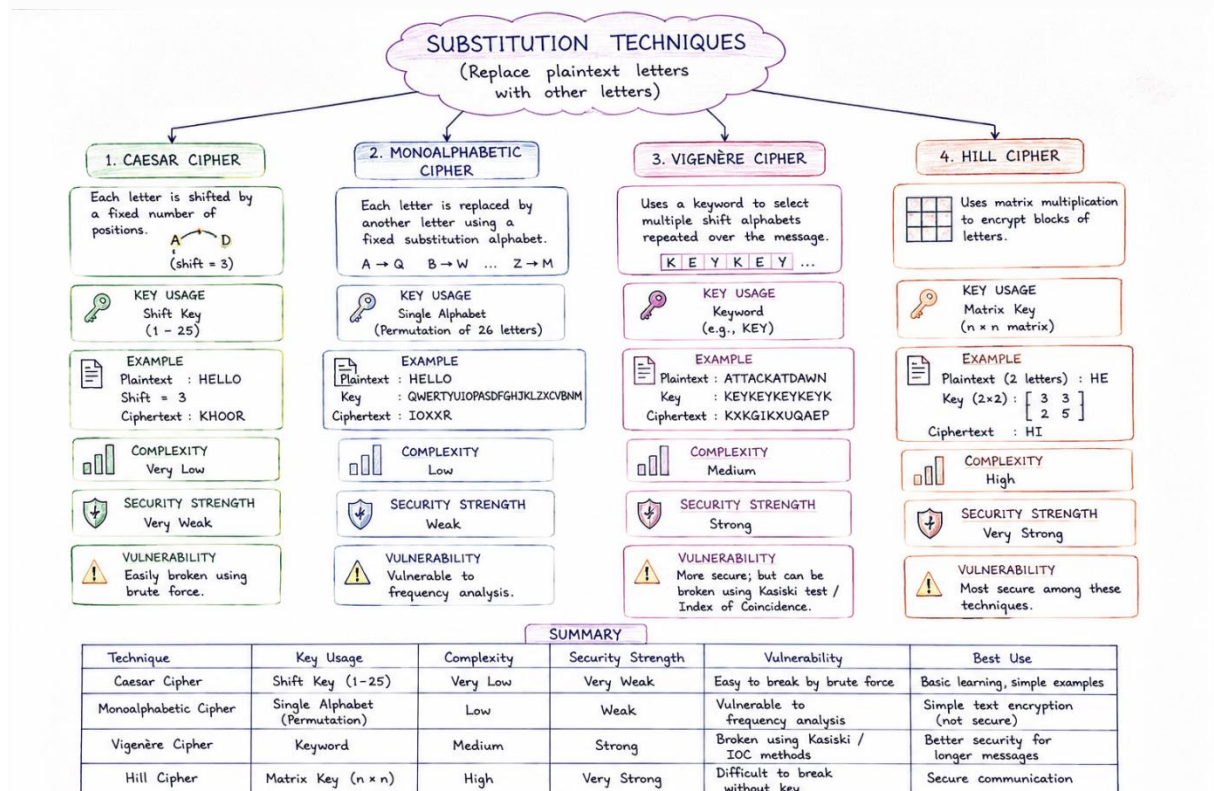
Introduction

Substitution ciphers are classical encryption techniques in which each letter of the plaintext is replaced with another letter or symbol based on a specific rule or key. Different substitution techniques provide different levels of security depending on key usage and encryption complexity.

There are different types of substitution techniques, such as **Caesar Cipher**, **Monoalphabetic Cipher**, **Vigenère Cipher**, and **Hill Cipher**, each offering different levels of security, key usage, and complexity. Some methods are simple and easy to implement but provide limited security, while others use more advanced mathematical operations to offer stronger protection against cryptographic attacks.

Comparing these techniques helps us understand their advantages, limitations, and security strength. It also enables us to identify the most suitable encryption method for secure communication based on the required level of data protection.

CONCEPT MAP



Comparison of Substitution Techniques

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Shift Key	Low	Very Weak
Monoalphabetic Cipher	Single Alphabet Key	Low	Weak
Vigenère Cipher	Keyword	Medium	Strong
Hill Cipher	Matrix Key	High	Very Strong

Analysis

- **Caesar Cipher** is the simplest technique and can be broken using brute force.
- **Monoalphabetic Cipher** is more secure than Caesar but is vulnerable to frequency analysis.

- **Vigenère Cipher** uses multiple alphabets, making it harder to crack.
- **Hill Cipher** uses matrix multiplication and provides the highest security among these techniques.
- **Which Technique is Most Secure and Why?**
- **Hill Cipher** is the most secure because:

- It uses matrix-based encryption.
- Multiple letters are encrypted together.
- It is difficult to break without the correct matrix key.

Conclusion

Among the four substitution techniques, **Hill Cipher** offers the highest security due to its mathematical approach and complex key structure. **Vigenère Cipher** provides moderate security, while **Caesar** and **Monoalphabetic Ciphers** are suitable only for basic learning purposes.

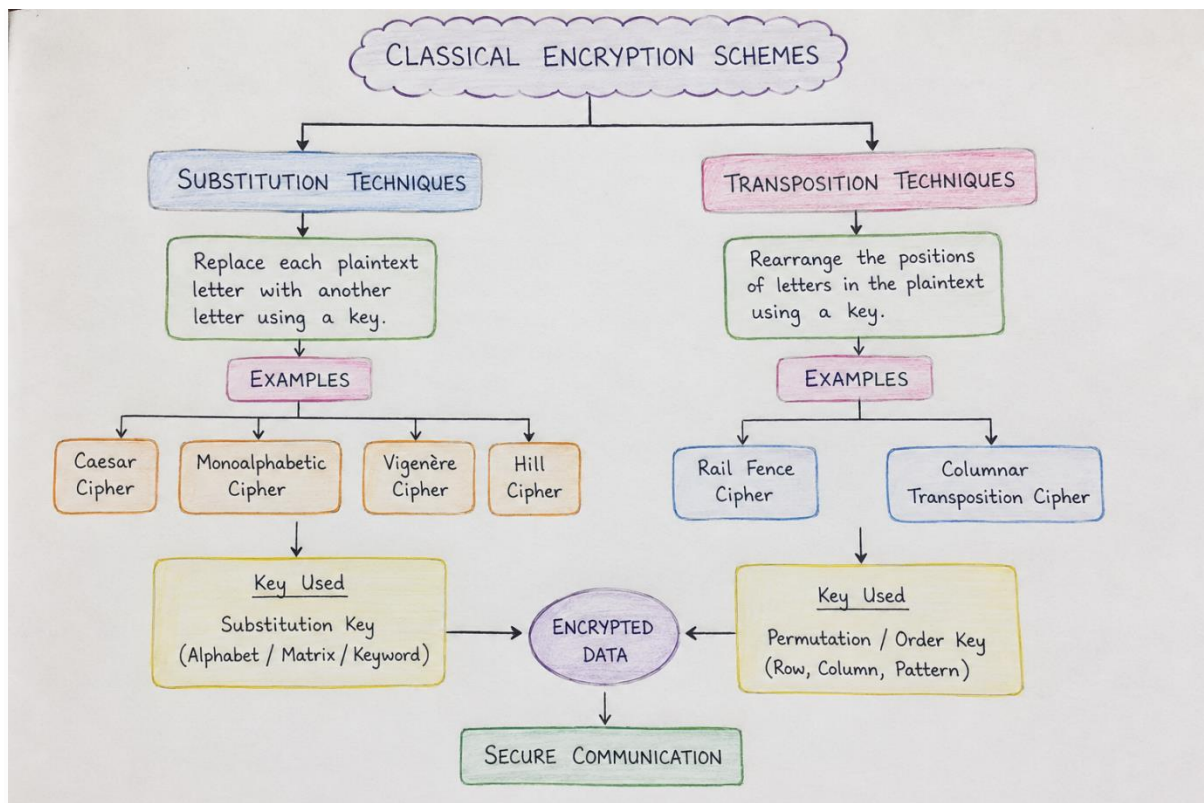
Question 3

Construct a Concept Map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques

Introduction

Classical encryption schemes are traditional methods used to secure information by converting plaintext into ciphertext. The two main categories are **Substitution Techniques** and **Transposition Techniques**. In substitution techniques, characters are replaced with other characters based on a key, while in transposition techniques, the positions of characters are rearranged without changing the characters themselves. Comparing these two techniques helps us understand how they affect data structure, security level, and resistance to cryptographic attacks.

Concept Map



Explanation of the Concept Map

1. Classical Encryption Schemes

Classical encryption schemes are the earliest methods of protecting information. They convert readable text (plaintext) into unreadable text (ciphertext) using encryption algorithms and keys.

They are divided into two major categories:

- **Substitution Techniques**
- **Transposition Techniques**

2. Substitution Techniques

In substitution techniques, each character in the plaintext is replaced with another character according to a specific rule or key.

Examples:

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

Features

- Changes the original letters.
- Easy to understand and implement.
- Security depends on the key used.

3. Transposition Techniques

In transposition techniques, the characters remain the same, but their positions are rearranged according to a key.

Examples:

- Rail Fence Cipher
- Columnar Transposition Cipher

Features

- Does not change the letters.
- Changes only the order of characters.
- Makes the message difficult to understand.

Task (a): Show Differences and Similarities

Differences

Substitution Technique	Transposition Technique
Replaces letters with different letters	Rearranges the positions of letters
Characters are changed	Characters remain the same
Uses substitution key	Uses permutation/order key
Example: Caesar Cipher	Example: Rail Fence Cipher

Similarities

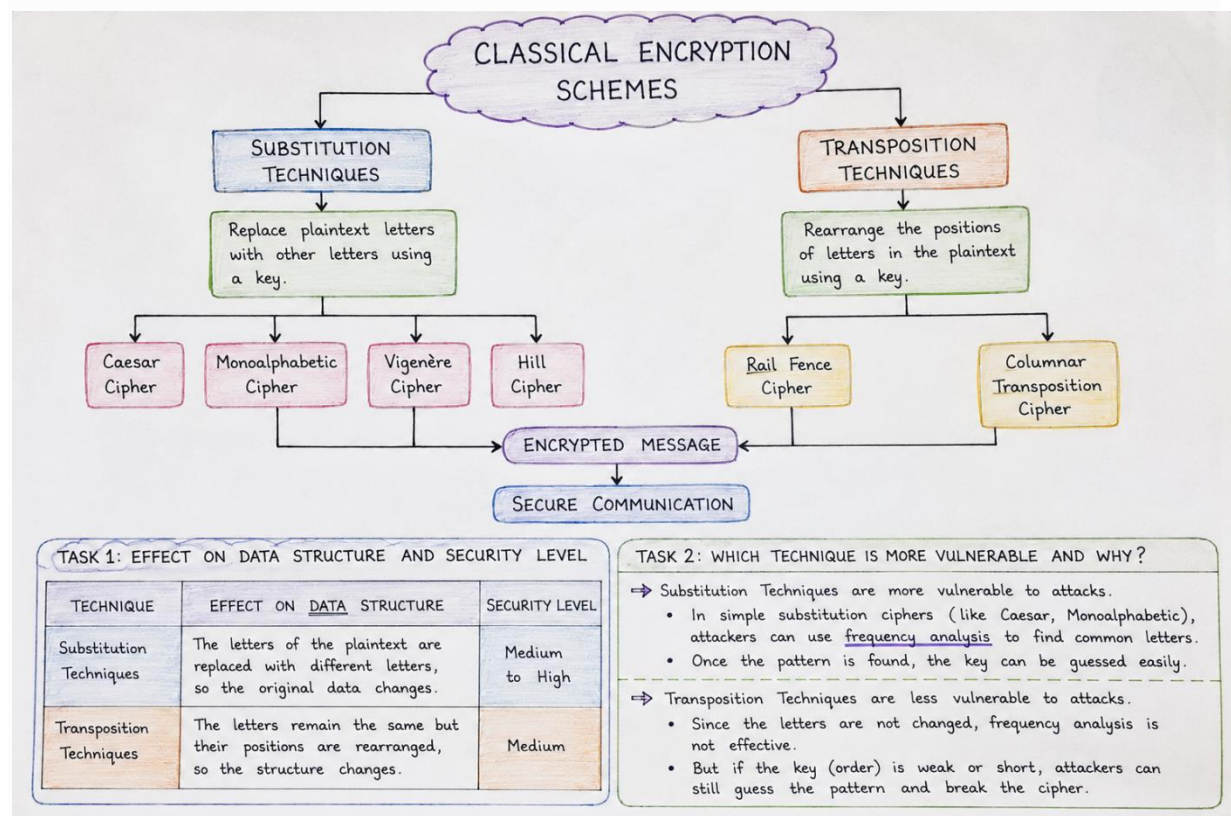
- Both are classical encryption methods.
- Both convert plaintext into ciphertext.
- Both require a key for encryption and decryption.
- Both improve data confidentiality.
- Both are used for secure communication.

Task (b)(i): Map how each affects the Data Structure and Security Level

Technique	Effect on Data Structure	Security Level
Substitution	Characters are replaced with different characters	Medium to High
Transposition	Character positions are rearranged while letters remain unchanged	Medium

Task (b)(ii): Analyze which is more vulnerable to attacks and why

Substitution techniques are generally more vulnerable to attacks because repeated letter patterns can be identified using **frequency analysis**, especially in simple ciphers like Caesar and Monoalphabetic. **Transposition techniques** are comparatively less vulnerable to frequency analysis because the letters remain the same but are rearranged. However, if the key is weak or short, attackers can still recover the original message.



Analysis

Both substitution and transposition techniques play an important role in classical cryptography. Substitution techniques focus on replacing characters, whereas transposition techniques focus on rearranging character positions. Individually, they provide moderate security, but combining both techniques results in stronger encryption and better protection against cryptographic attacks.

Conclusion

Classical encryption schemes mainly consist of substitution and transposition techniques. While substitution changes the characters and transposition changes their positions, both aim to secure sensitive information. Combining these techniques provides better security than using either method alone, making encrypted communication more resistant to attacks.

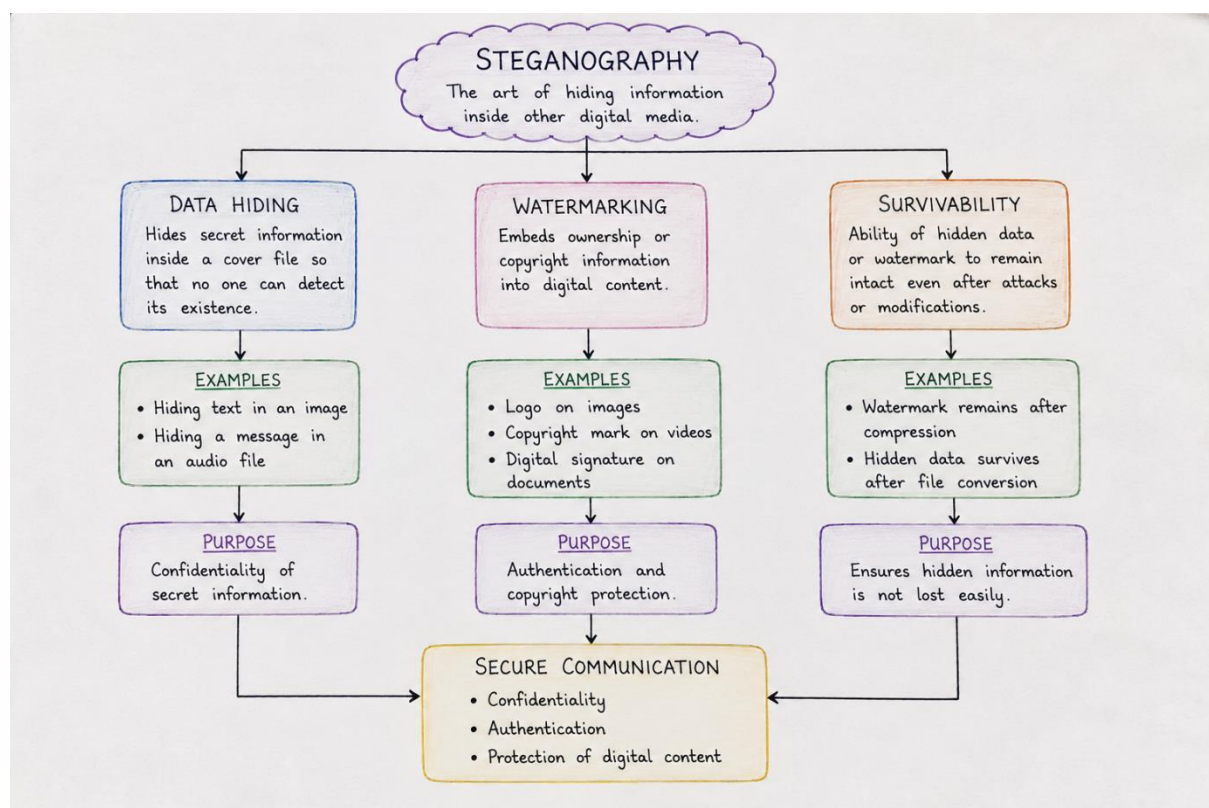
Question 4

Create a Concept Map Connecting Steganography Concepts: Data Hiding, Watermarking and Survivability

Introduction

Steganography is the practice of hiding secret information inside digital media such as images, audio, videos, or text so that the existence of the message remains unknown. It is widely used for secure communication, copyright protection, and information security. The main concepts related to steganography are **Data Hiding**, **Watermarking**, and **Survivability**, which work together to ensure that hidden information remains secure, authentic, and resistant to attacks.

CONCEPT MAP



Explanation of the Concept Map

1. Steganography

Steganography is a security technique that hides secret information inside a cover file such as an image, audio, or video without attracting attention.

Purpose

- Hide confidential information.
- Enable secure communication.
- Protect sensitive digital data.

2. Data Hiding

Data hiding is the process of embedding secret information inside another digital file so that it cannot be noticed by unauthorized users.

Examples

- Hiding text inside an image.
- Hiding a message inside an audio file.

3. Watermarking

Watermarking embeds ownership or copyright information into digital content.

Examples

- Company logos on images.
- Copyright marks on videos.
- Digital signatures on documents.

4. Survivability

Survivability refers to the ability of the hidden data or watermark to remain intact even after compression, editing, resizing, or transmission.

Examples

- Watermark remains visible after image compression.
- Hidden information survives file conversion.

Task (a): Show Relationships Between These Concepts

Concept	Relationship
Steganography → Data Hiding	Hides secret information inside a cover file.
Data Hiding → Watermarking	Watermarking is a special application of data hiding.
Watermarking → Survivability	Watermarks should remain even after modifications.
Survivability → Secure Communication Ensures hidden data remains protected and recoverable.	

Task (b): Analyze How Each Contributes to Secure Communication

Concept	Contribution
Data Hiding	Protects confidential information from unauthorized users.
Watermarking	Verifies ownership and prevents copyright violations.
Survivability	Maintains hidden data even after attacks or file modifications.

Task (c): One Real-World Application for Each

Concept	Real-World Application
Data Hiding	Secret military communication using hidden messages in images.
Watermarking	Copyright protection of digital photographs and videos.
Survivability	Secure medical images where hidden patient information remains after compression.

Analysis

Steganography provides secure communication by hiding confidential information inside digital media. Data hiding conceals the information, watermarking protects ownership, and survivability ensures that the hidden information remains available even after modifications or attacks. Together, these concepts improve data security and authenticity.

Conclusion

Steganography, data hiding, watermarking, and survivability are closely related concepts in information security. Data hiding keeps information confidential, watermarking protects ownership, and survivability ensures the hidden information remains intact. Together, they provide secure and reliable communication while protecting digital content from unauthorized access and tampering.

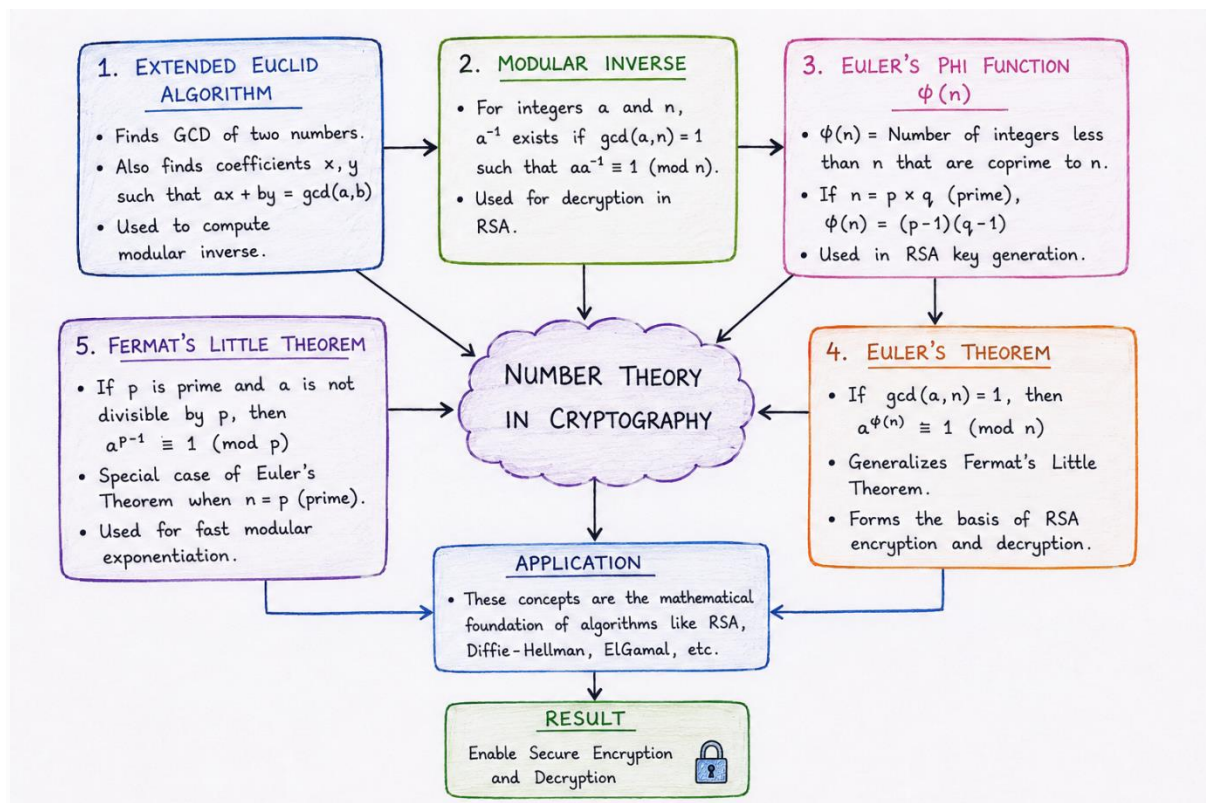
Question 5

Draw a Concept Map Linking Number Theory Concepts Used in Cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem

Introduction

Number theory is the mathematical foundation of modern cryptography. Many public-key encryption algorithms, such as RSA, depend on number theory concepts to generate keys, perform encryption and decryption, and ensure secure communication. Important concepts include the **Extended Euclidean Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem**, and **Fermat's Little Theorem**. These concepts work together to provide secure and efficient cryptographic operations.

Concept Map



Explanation of the Concept Map

1. Extended Euclidean Algorithm

The Extended Euclidean Algorithm is used to calculate the Greatest Common Divisor (GCD) of two numbers and also finds the **modular inverse**.

Purpose

- Finds GCD
- Calculates modular inverse
- Used in RSA key generation

2. Modular Inverse

A modular inverse is a number that satisfies $a \times a^{-1} \equiv 1 \pmod{n}$

Purpose

- Used during RSA decryption
- Generates private keys
- Performs modular arithmetic

Euler's Phi Function

Euler's Phi Function ($\phi(n)$) gives the number of positive integers less than n that are relatively prime to n .

Purpose

- Used in RSA algorithm
- Calculates encryption and decryption keys

4. Euler's Theorem

Euler's Theorem states that $a^{\phi(n)} \equiv 1 \pmod{n}$ when $\gcd(a,n)=1$.

Purpose

- Forms the mathematical basis of RSA
- Helps in secure encryption and decryption

5. Fermat's Little Theorem

Fermat's Little Theorem states that $a^{p-1} \equiv 1 \pmod{p}$ where p is a prime number.

Purpose

- Simplifies modular arithmetic
- Used in cryptographic calculations
- Basis for several public-key algorithms

Task (a): Show how each concept is mathematically connected

Concept	Mathematical Connection
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Concept	Mathematical Connection
Extended Euclidean Algorithm	Finds GCD and Modular Inverse
Modular Inverse	Used with Euler's Phi Function
Euler's Phi Function	Used in Euler's Theorem
Euler's Theorem	Basis of RSA encryption
Fermat's Little Theorem	Special case of Euler's Theorem for prime numbers

Task (b): Map their role in Encryption/Decryption

Concept	Role
Extended Euclidean Algorithm	Generates private key
Modular Inverse	Performs decryption
Euler's Phi Function	Generates RSA keys
Euler's Theorem	Encrypts and decrypts messages
Fermat's Little Theorem	Speeds modular exponentiation

Task (c): Which concept is most critical for Public-Key Cryptography? Justify

Euler's Theorem is the most critical concept because it forms the mathematical foundation of the **RSA cryptosystem**, which is one of the most widely used public-key encryption algorithms. It ensures that encrypted messages can be decrypted correctly using mathematically related public and private keys.

Analysis

All number theory concepts work together in modern cryptography. The Extended Euclidean Algorithm calculates the modular inverse, Euler's Phi Function determines key values, Euler's Theorem provides the mathematical basis for encryption and decryption, and Fermat's Little Theorem simplifies modular calculations. Together, these concepts ensure secure and efficient public-key cryptography.

Conclusion

Number theory plays a vital role in cryptography by providing the mathematical principles required for secure communication. Concepts such as the Extended Euclidean Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem, and Fermat's Little Theorem are interconnected and enable secure encryption, decryption, and key generation. Among them, **Euler's Theorem** is the most important because it serves as the foundation of RSA public-key cryptography.

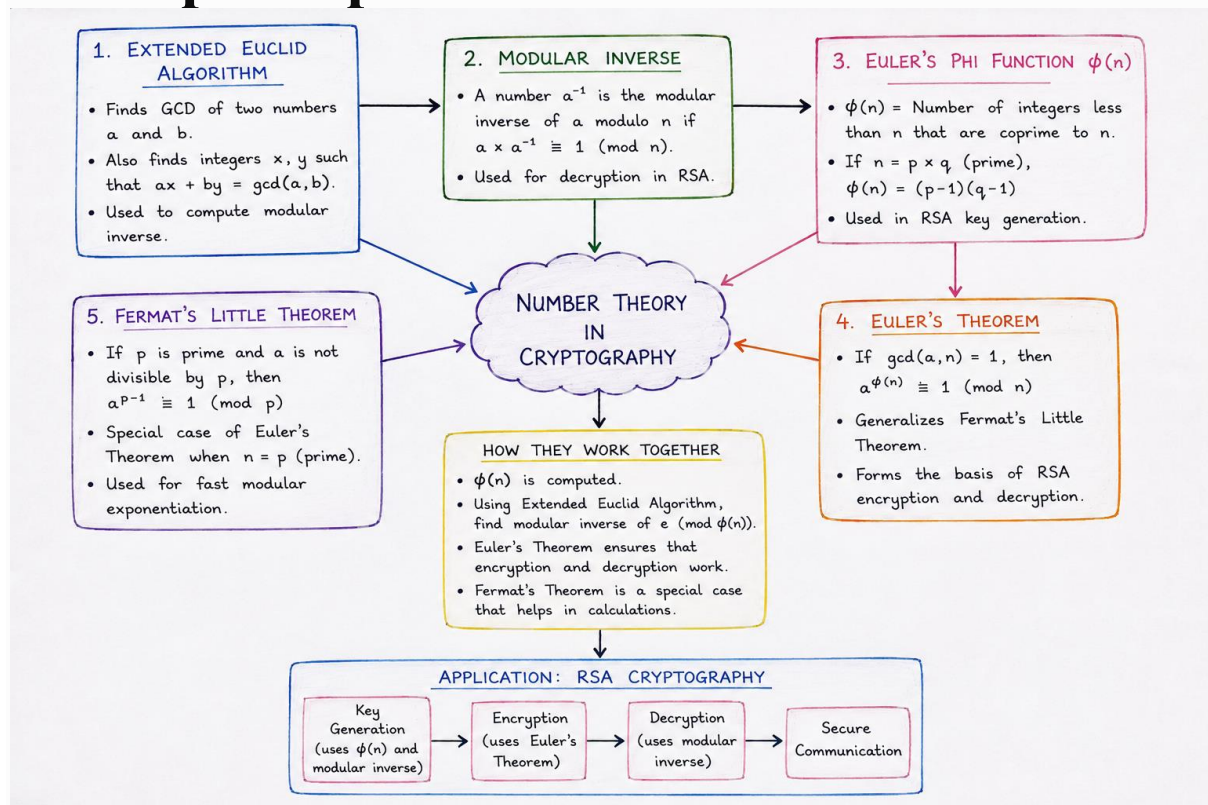
Question 5

Draw a Concept Map Linking Number Theory Concepts Used in Cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem

Introduction

Number theory is the foundation of modern cryptography. It provides the mathematical concepts required for secure encryption, decryption, and key generation. Algorithms such as RSA use concepts like the **Extended Euclidean Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem**, and **Fermat's Little Theorem** to protect sensitive information. These concepts are interconnected and play an important role in ensuring secure communication.

Concept Map



Explanation of the Concept Map

1. Extended Euclidean Algorithm

The Extended Euclidean Algorithm is used to find the **Greatest Common Divisor (GCD)** of two numbers. It also calculates the **modular inverse**, which is essential for public-key cryptography.

Purpose

- Finds GCD of two numbers.
- Computes modular inverse.
- Used during RSA key generation.

2. Modular Inverse

A modular inverse is a number that satisfies:

$$a \times a^{-1} \equiv 1 \pmod{n}$$

It is mainly used in cryptographic algorithms for decryption.

Purpose

- Used to generate private keys.
- Performs decryption in RSA.
- Helps in modular arithmetic operations.

3. Euler's Phi Function

Euler's Phi Function, denoted as $\phi(n)$, gives the number of positive integers less than n that are relatively prime to n .

Purpose

- Calculates the value of $\phi(n)$.
- Used in RSA key generation.
- Helps generate encryption and decryption keys.

4. Euler's Theorem

Euler's Theorem states that if $\gcd(a, n) = 1$, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This theorem forms the mathematical basis of RSA encryption.

Purpose

- Basis of RSA cryptography.
- Used in encryption and decryption.
- Ensures secure communication.

5. Fermat's Little Theorem

Fermat's Little Theorem states that if p is a prime number, then:

$$a^{(p-1)} \equiv 1 \pmod{p}$$

It is a special case of Euler's Theorem.

Purpose

- Simplifies modular arithmetic.
- Used in cryptographic computations.
- Helps perform faster calculations.

Task (a): Show how each concept is mathematically connected

Concept	Mathematical Connection
Extended Euclidean Algorithm	Finds GCD and calculates Modular Inverse.
Modular Inverse	Used with Euler's Phi Function to generate keys.
Euler's Phi Function	Used in Euler's Theorem.
Euler's Theorem	Mathematical foundation of RSA encryption.
Fermat's Little Theorem	Special case of Euler's Theorem when n is prime.

Task (b): Map their role in Encryption and Decryption

Concept	Role in Cryptography
Extended Euclidean Algorithm	Finds modular inverse for private key generation.
Modular Inverse	Used during RSA decryption.
Euler's Phi Function	Generates public and private keys.
Euler's Theorem	Performs secure encryption and decryption.
Fermat's Little Theorem	Speeds up modular exponentiation calculations.

Task (c): Analyze which concept is most critical for Public-Key Cryptography and Justify

Euler's Theorem is the most critical concept because it provides the mathematical foundation for the RSA algorithm, one of the most widely used public-key cryptographic systems. It guarantees that encrypted messages can be decrypted correctly using mathematically related public and private keys. Without Euler's Theorem, RSA encryption and decryption would not work efficiently.

Analysis

The concepts of number theory are closely connected in cryptography. The Extended Euclidean Algorithm computes the modular inverse, Euler's Phi Function generates key values, Euler's Theorem enables secure encryption and decryption, and Fermat's Little Theorem simplifies modular computations. Together, these concepts make modern public-key cryptography secure, reliable, and efficient.

Conclusion

Number theory concepts are essential for secure cryptographic systems. The Extended Euclidean Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem, and Fermat's Little Theorem work together to support key generation, encryption, and decryption. Among them, **Euler's Theorem** is the most important because it forms the mathematical basis of RSA, ensuring secure communication over digital networks.